

Class 10: Structural Bioinformatics 1

Christeena Pano (PID: A17842497)

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

Introduction to the RCSB Protein Data Bank (PDB)

The PDB archive is the major repository of information about the 3D structures of large biological molecules, including proteins and nucleic acids. Understanding the shape of these molecules helps to understand how they work. This knowledge can be used to help deduce a structure's role in human health and disease, and in drug development. The structures in the PDB range from tiny proteins and bits of DNA or RNA to complex molecular machines like the ribosome composed of many chains of protein and RNA.

In the first section of this lab we will interact with the main US based PDB website.

PDB Statistics and Data Import

What is in the PDB in terms of molecule type and structure determination method.

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

The percentage of structures in the PDB solved by X-Ray is 80.48577%. The percentage of structures in the PDB that are solved by Electron Microscopy is 13.38046%.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

Two functions that can help here are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {  
  tmp <- gsub(",", "", x)  
  sum(as.numeric(tmp))  
}
```

```
n.tot <- rm.comma(pdb$Total)  
n.xray <- rm.comma(pdb$X.ray)  
n.em <- rm.comma(pdb$EM)  
  
n.xray/n.tot *100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr)  
  
read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
  `Molecular Type`      `X-ray`      EM      NMR Integrative `Multiple methods` Neutron
  <chr>                <dbl> <dbl> <dbl>          <dbl>          <dbl>
1 Protein (only)      180758 23111 12813          348            229      84
2 Protein/Oligosacch~  10488  3741   34           8             11       1
3 Protein/NA           9205  6751  287          26             8       0
4 Nucleic acid (only)  3154   250 1578           3            15       3
5 Other                178    27   35           4             0       0
6 Oligosaccharide (o~   11     0    6           0             1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

Q2: What proportion of structures in the PDB are protein?

```
pdb$Total[1]
```

```
[1] "217,375"
```

The total number of protein sequences is 202,556,314, according to Uniprot database.

```
(217375/202556314) *100
```

```
[1] 0.1073158
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography)

```
n.tot <- rm.comma(pdb$Total)
n.pro <- rm.comma(pdb$`Protein (only)`) +
  rm.comma(pdb$`Protein/Oligosaccharide`) +
  rm.comma(pdb$`Protein/NA`)
round(sum(n.pro) / sum(n.tot) * 100, 2)
```

```
[1] 0
```

```
n.tot <- rm.comma(pdb$Total)
n.pro <- rm.comma(pdb$Total[1]) # row 1 = Protein (only)
round(n.pro / n.tot * 100, 2)
```

```
[1] 85.85
```

```
n.tot <- rm.comma(pdb$Total)
n.pro <- rm.comma(pdb$Total[1:3])

round(n.pro / n.tot * 100, 2)
```

[1] 97.92

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1227 HIV-1 protease structures in the current PDB.

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at <https://molstar.org/viewer/>.

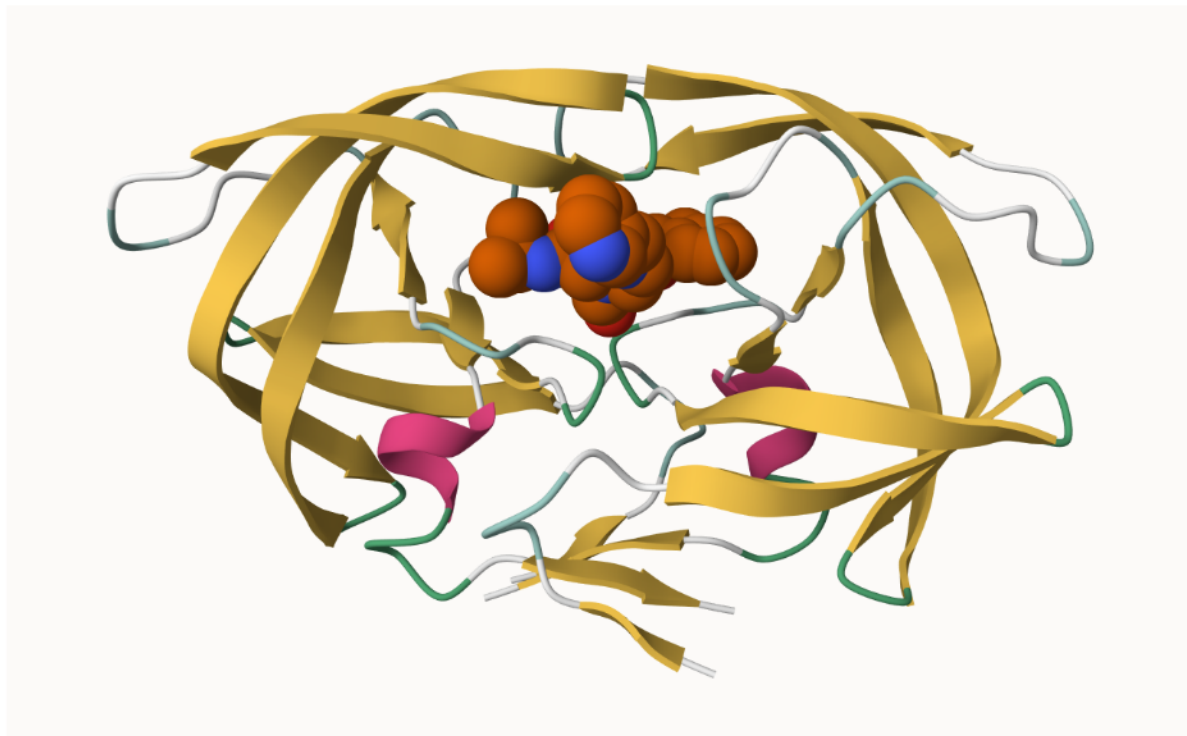


Figure 1: The HIV-protease enzyme is a homodimer with two chains



Figure 2: The HIV-protease enzyme is a homodimer with two chains

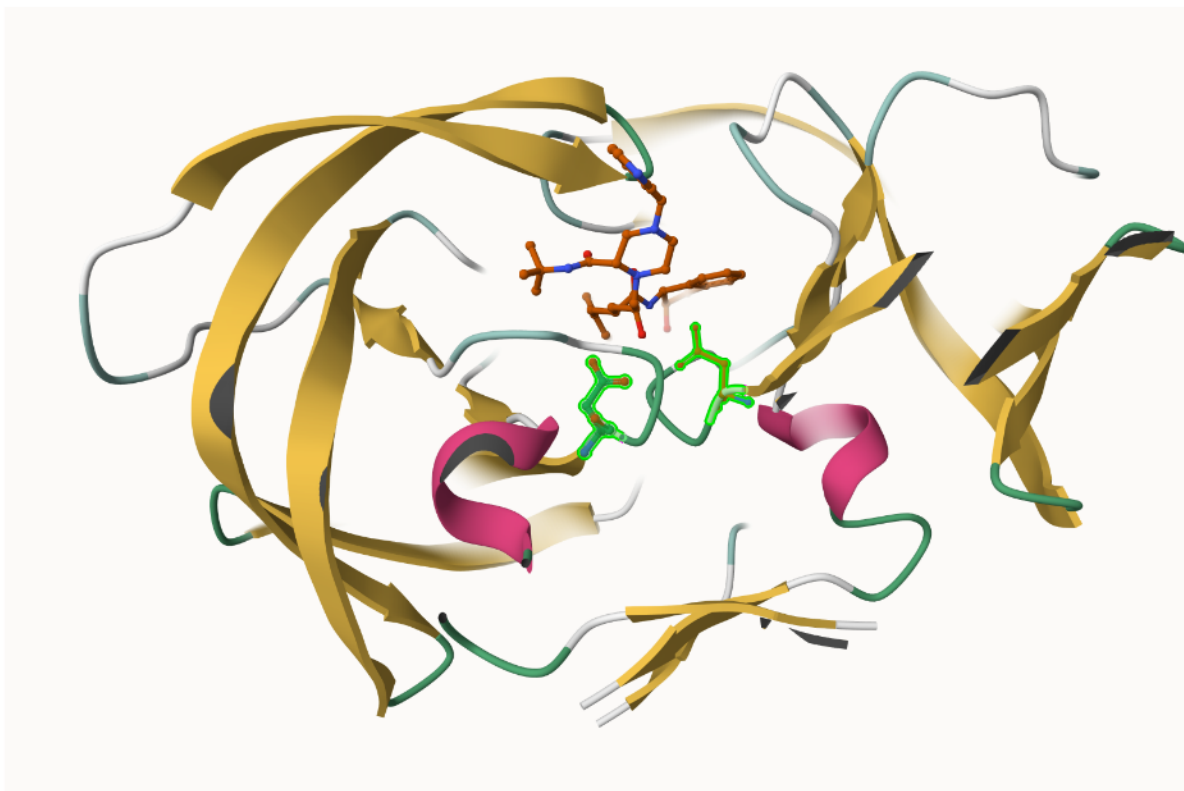


Figure 3: The HIV-protease enzyme with both Asp 25 positions selected

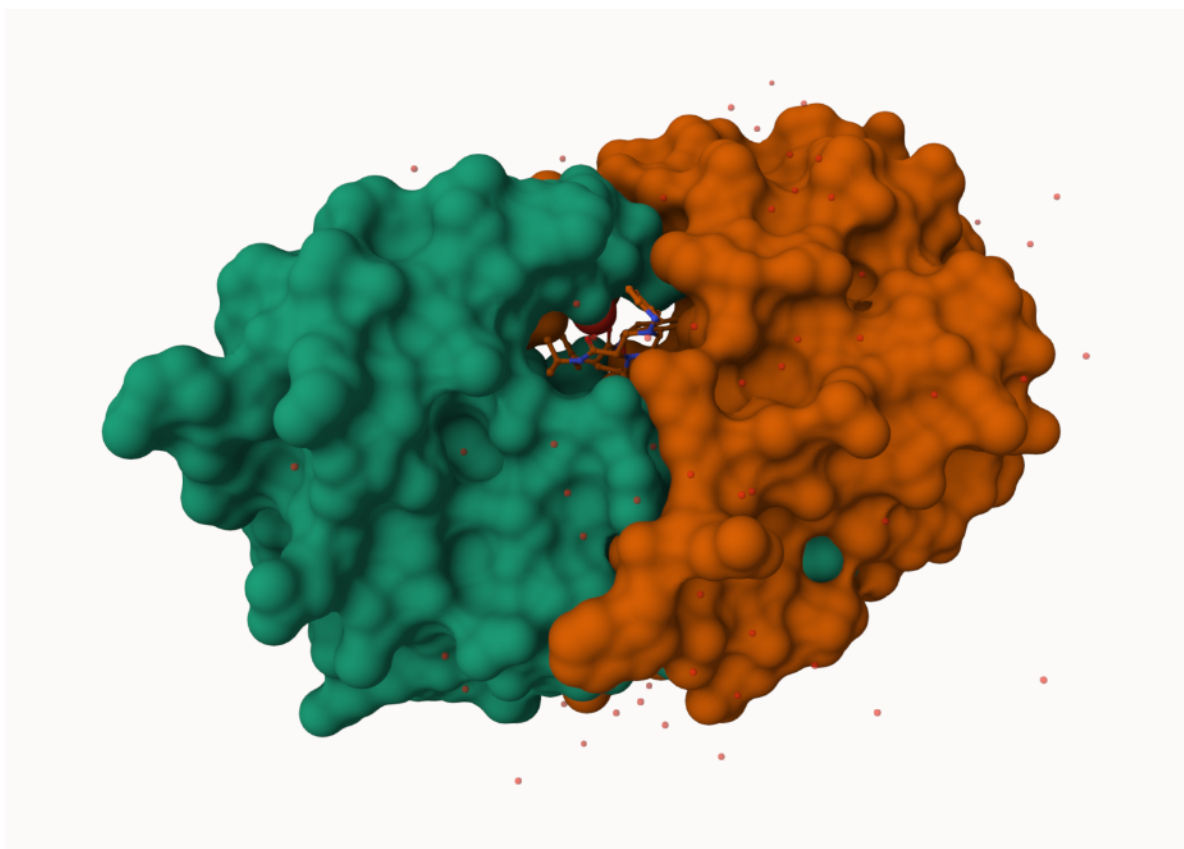


Figure 4: HIV-Pr molecular surface representation display of polymer showing the binding cleft site for the inhibitor (drug) molecule

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Since hydrogen only has one electron, it is not visible with the methods used to solve/find these structures. Electron microscopy and X-ray diffraction better detect the oxygen molecule. Therefore, only one atom (oxygen) is used to represent the water molecules in this structure.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

This water molecule is HOH 308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

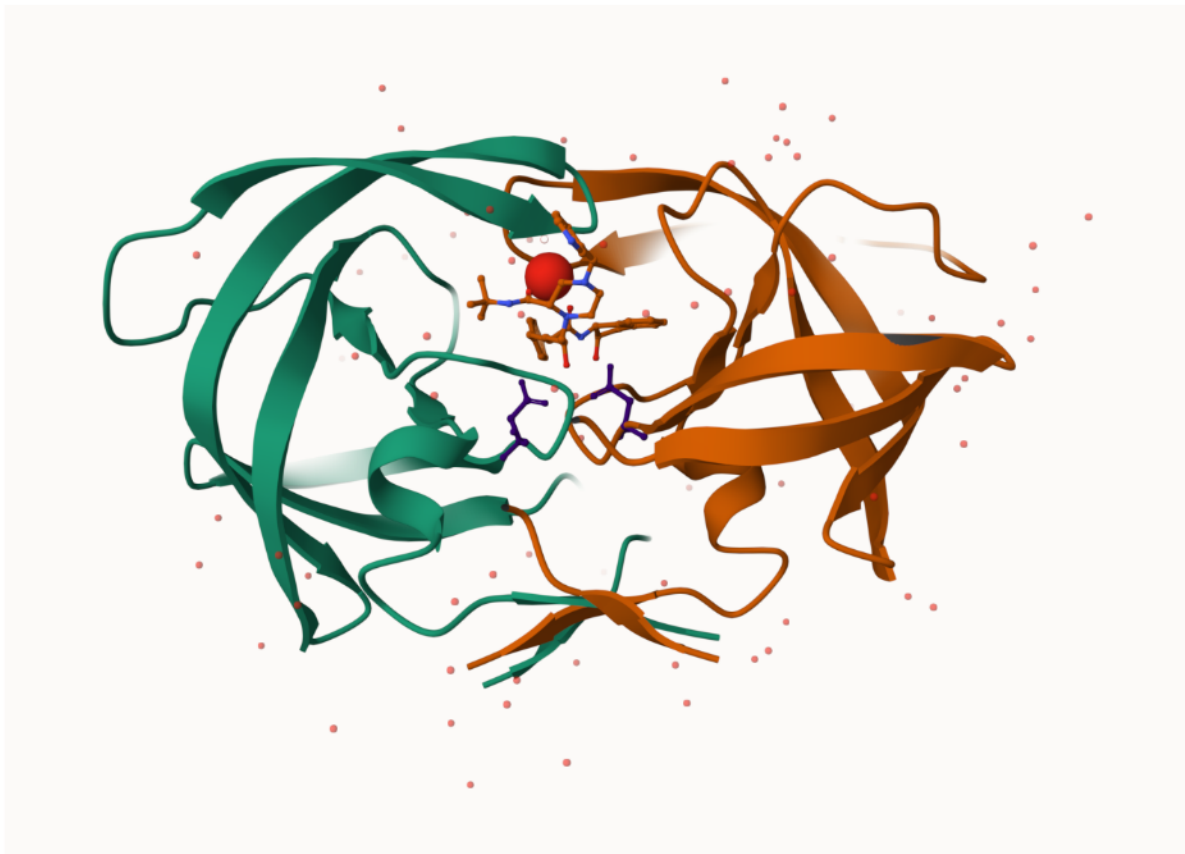


Figure 5: HIV-protease cartoon polymer with both chains A and B, the ligand, both Asp 25 residues selected, and the critical water molecule

```
# Pak can install from multiple sources
#install.packages("pak")

#| eval: !expr knitr::is_html_output()

pak::pkg_install(c("bioboot/bio3dview",
  "NGLVieweR",
  "bioc::msa"))
```

! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds\_set()`.

Loading metadata database

Loading metadata database ... done

No downloads are needed

3 pkgs + 47 deps: kept 46 [35.7s]

```
#install.packages("bio3d")  
library(bio3d)
```

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)  
  
pdb <- read.pdb("1hsg")  
pdb
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this pdb object.

Q8: Name one of the two non-protein residues?

MK1 is one of the two non-protein residues.

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure.

```
head(pdb$atom)
```

NULL

There are lots of functions that can work with these `pdb` objects:

```
#head( pdbseq(pdb))
```

```
#| eval: !expr knitr::is_html_output()
```

```
library(bio3dview)
```

```
library(NGLVieweR)
```

```
#| eval: !expr knitr::is_html_output()
```

```
#view.pdb(pdb) |> setSpin()
```

```
#install.packages("remotes")
```

```
remotes::install_github("bioboot/bio3dview")
```

Skipping install of 'bio3dview' from a github remote, the SHA1 (568c6e99) has not changed since last install.
Use `force = TRUE` to force installation

```
#install.packages("NGLVieweR")
```

let's try a custom view

```
#view.pdb(pdb,colorScheme="sse", backgroundColor="black")
```

Q. Create a custom view highlighting the active site ASP (resno25), the two chains (in your choice of colors) and the ligand all on a custom color background

```
library(NGLVieweR)
```

```
active.site <- atom.select(pdb, resno=25)
```

```
#| eval: !expr knitr::is_html_output()
```

```
#view.pdb(pdb,  
  cols= c("red", "blue"),  
  highlight = active.site,  
  highlight.style = "spacefill",  
  backgroundColor= "pink") |>  
setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

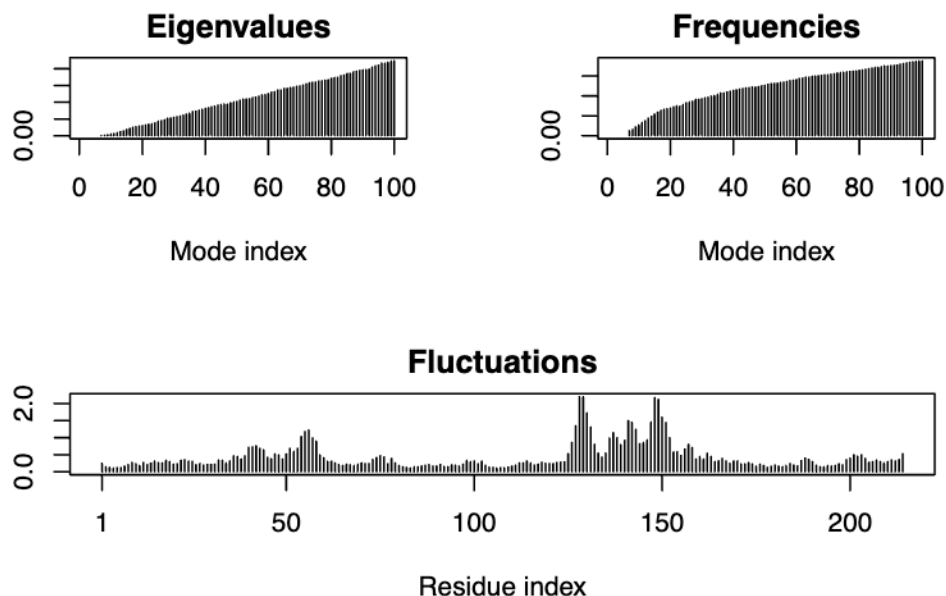
```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
m <- nma( adk )
```

```
Building Hessian...      Done in 0.012 seconds.
Diagonalizing Hessian... Done in 0.277 seconds.
```

```
plot(m)
```



View the results with an interactive structure view

```
view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa is found only on BioConductor.

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dview is not found on Bioconductor or CRAN.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

This is true.

Our first step is find a sequence for this family. We will use the database ID "1ake_A" here:

```
id <- "1ake_A"
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTRKDDQEETVRKRLVEYHQMTPALIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = 02NT9BV9014

.....

Reporting 96 hits

```
head(blast$hit.tbl)
```

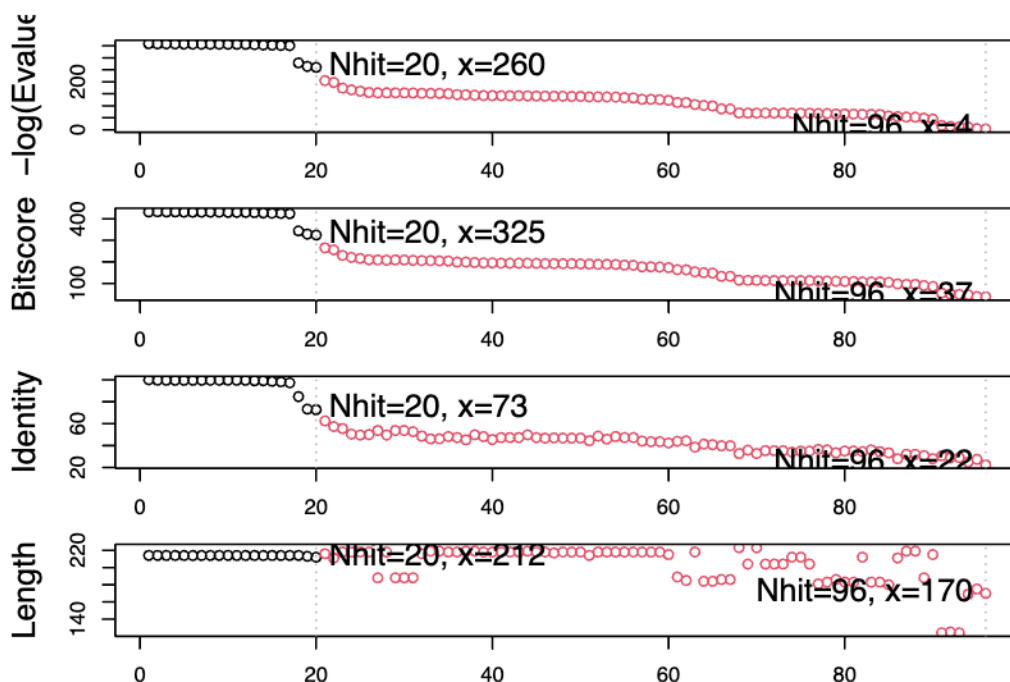
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_2916143	1AKE_A	100.000	214	0	0	1
2	Query_2916143	8BQF_A	99.533	214	1	0	1
3	Query_2916143	4X8M_A	99.533	214	1	0	1
4	Query_2916143	6S36_A	99.533	214	1	0	1
5	Query_2916143	9R6U_A	99.533	214	1	0	1
6	Query_2916143	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evaluate	pdb.id	acc
1	214	1	214	1.82e-156	432	100.00	358.6044	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.77e-156	432	100.00	357.6409	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits <- plot(blast)
```

```
* Possible cutoff values:    260 3
    Yielding Nhits:         20 96

* Chosen cutoff value of:    260
    Yielding Nhits:         20
```



```
hits$ pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
# Download related PDB files
```

```
files <- get.pdb(hits$ pdb.id, path="pdds", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdds", split = TRUE, gzip = TRUE):
pdds/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdds", split = TRUE, gzip = TRUE):
pdds/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdds", split = TRUE, gzip = TRUE):
pdds/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdds", split = TRUE, gzip = TRUE):
pdds/6S36.pdb.gz exists. Skipping download
```


Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

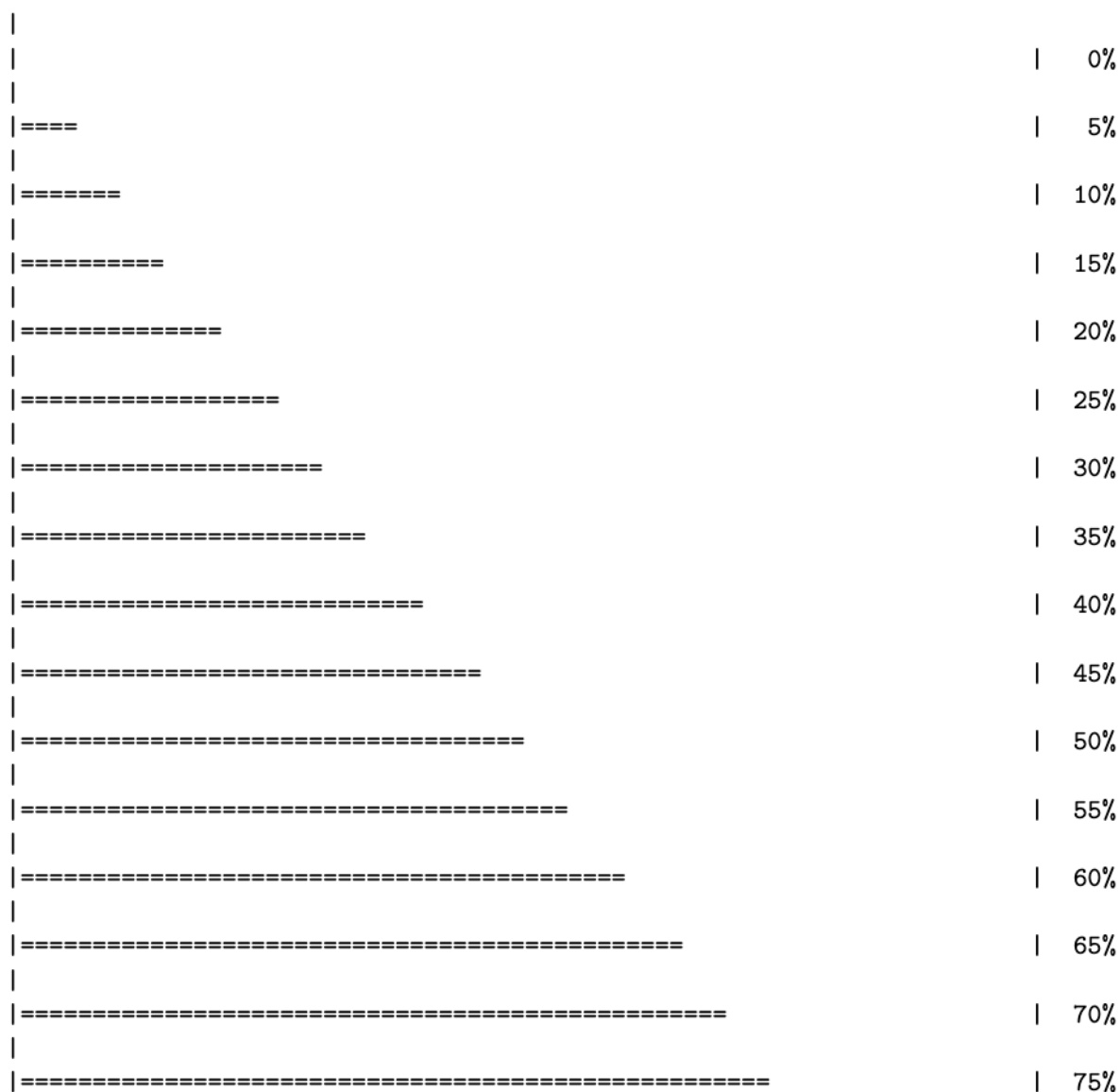
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download



```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

```

# Align related PDBs
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")

```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE

```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbs/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbs/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbs/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbs/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbs/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20  name: pdbs/split_chain/4NP6_A.pdb

```

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA				
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMLRAA				
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMLRAA				
	***** ***** *^*****				
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---				
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR				
[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR				

[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVTDDELVIALVRERICQEDSR	
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVTDDEIIIALVKERICQEDSR	
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVTDDELVIALVKERIAQEDCR	
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA	
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE	
	^* *~*** ** ^~* **~*~^~^*~^*** *	
	41 . . . 80	
	81 . . . 120	
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD	
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD	
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE	
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE	
	**** *****^~** *^ *****^***** * ^**^	
	81 . . . 120	
	121 . . . 160	
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA	
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	

[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRE
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRAHLPSGRTYHVYNPPKVEGKDDVTGEDLVIRE
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:4] 6S36_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb	TKPVAEVRADLEKILG

```

[Truncated_Name:6] 9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structure view

```
#view.pdb(pdb)
```

```
# Vector containing PDB database codes
```

```
ids <- basename.pdb(pdb$id)
```

```
anno <- pdb.annotate(ids)
```

```
unique(anno$source)
```

```
[1] "Escherichia coli"
```

```
[2] "Escherichia coli K-12"
```


- [3] "Escherichia coli 0139:H28 str. E24377A"
- [4] "Escherichia coli str. K-12 substr. MDS42"
- [5] "Photobacterium profundum"
- [6] "Vibrio cholerae 01 biovar El Tor str. N16961"

anno

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
8BQF_A	8BQF	A	Protein	234	X-
ray					
4X8M_A	4X8M	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
9R6U_A	9R6U	A	Protein	214	X-
ray					
9R71_A	9R71	A	Protein	214	X-
ray					
8Q2B_A	8Q2B	A	Protein	214	X-
ray					
8RJ9_A	8RJ9	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
4X8H_A	4X8H	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					

8PVW_A	8PVW	A	Protein	187	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
4NP6_A	4NP6	A	Protein	217	X-
ray					

	resolution	scopDomain	pfam
1AKE_A	2.000	Adenylate kinase	Adenylate kinase (ADK)
8BQF_A	2.050	<NA>	Adenylate kinase (ADK)
4X8M_A	2.600	<NA>	Adenylate kinase (ADK)
6S36_A	1.600	<NA>	Adenylate kinase (ADK)
9R6U_A	1.770	<NA>	Adenylate kinase (ADK)
9R71_A	1.610	<NA>	Adenylate kinase (ADK)
8Q2B_A	1.760	<NA> Adenylate kinase, active site lid (ADK_lid)	
8RJ9_A	1.590	<NA> Adenylate kinase, active site lid (ADK_lid)	
6RZE_A	1.690	<NA>	Adenylate kinase (ADK)
4X8H_A	2.500	<NA>	Adenylate kinase (ADK)
3HPR_A	2.000	<NA>	Adenylate kinase (ADK)
1E4V_A	1.850	Adenylate kinase	Adenylate kinase (ADK)
5EJE_A	1.900	<NA>	Adenylate kinase (ADK)
1E4Y_A	1.850	Adenylate kinase	Adenylate kinase (ADK)
3X2S_A	2.800	<NA>	<NA>
6HAP_A	2.700	<NA>	Adenylate kinase (ADK)
6HAM_A	2.550	<NA>	Adenylate kinase (ADK)
8PVW_A	2.000	<NA>	Adenylate kinase (ADK)
4K46_A	2.010	<NA>	Adenylate kinase (ADK)
4NP6_A	2.004	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
8BQF_A	AP5
4X8M_A	<NA>
6S36_A	CL (3),NA,MG (2)
9R6U_A	AP5,GOL,NA
9R71_A	AP5
8Q2B_A	AP5,S04,MPO
8RJ9_A	ADP (2)
6RZE_A	NA (3),CL (2)
4X8H_A	<NA>
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,C0
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG

6HAP_A	AP5	
6HAM_A	AP5	
8PVW_A	AP5	
4K46_A	ADP, AMP, PO4	
4NP6_A	<NA>	
		ligandName
1AKE_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
8BQF_A		BIS(ADENOSINE)-5'-
PENTAPHOSPHATE		
4X8M_A		<NA>
6S36_A	CHLORIDE ION (3), SODIUM ION, MAGNESIUM ION (2)	
9R6U_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE, GLYCEROL, SODIUM ION	
9R71_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
8Q2B_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE, SULFATE ION, 3[N-MORPHOLINO]PROPANE SULFONIC ACID	
8RJ9_A	ADENOSINE-5'-	
DIPHOSPHATE (2)		
6RZE_A	SODIUM ION (3), CHLORIDE ION (2)	
4X8H_A		<NA>
3HPR_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
1E4V_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE, COBALT (II) ION	
1E4Y_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2), BIS(ADENOSINE)-5'-PENTAPHOSPHATE, MAGNESIUM ION	
6HAP_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
6HAM_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
8PVW_A	BIS(ADENOSINE)-5'-	
PENTAPHOSPHATE		
4K46_A	ADENOSINE-5'-DIPHOSPHATE, ADENOSINE MONOPHOSPHATE, PHOSPHATE ION	
4NP6_A		<NA>
	source	
1AKE_A	Escherichia coli	
8BQF_A	Escherichia coli	
4X8M_A	Escherichia coli	
6S36_A	Escherichia coli	
9R6U_A	Escherichia coli	
9R71_A	Escherichia coli	

8Q2B_A	Escherichia coli
8RJ9_A	Escherichia coli
6RZE_A	Escherichia coli
4X8H_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
8PVW_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
4NP6_A	Vibrio cholerae 01 biovar El Tor str. N16961

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB

8BQF_A		
4X8M_A		
6S36_A		
9R6U_A		
9R71_A		
8Q2B_A	E. coli Adenylate Kinase variant D158A (	
8RJ9_A	E. coli adenylate kinase Asp84	
6RZE_A		
4X8H_A		
3HPR_A		
1E4V_A		
loop		
5EJE_A		
1E4Y_A		
loop		
3X2S_A		
conjugated adenylate kinase		
6HAP_A		
6HAM_A		
8PVW_A		
4K46_A		
4NP6_A		

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA	
8BQF_A	Scheerer, D., et al. Proc Natl Acad Sci U S A (2023)	0.22073	0.25789	
4X8M_A	Kovermann, M., et al. Nat Commun (2015)	0.24910	0.30890	
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560	

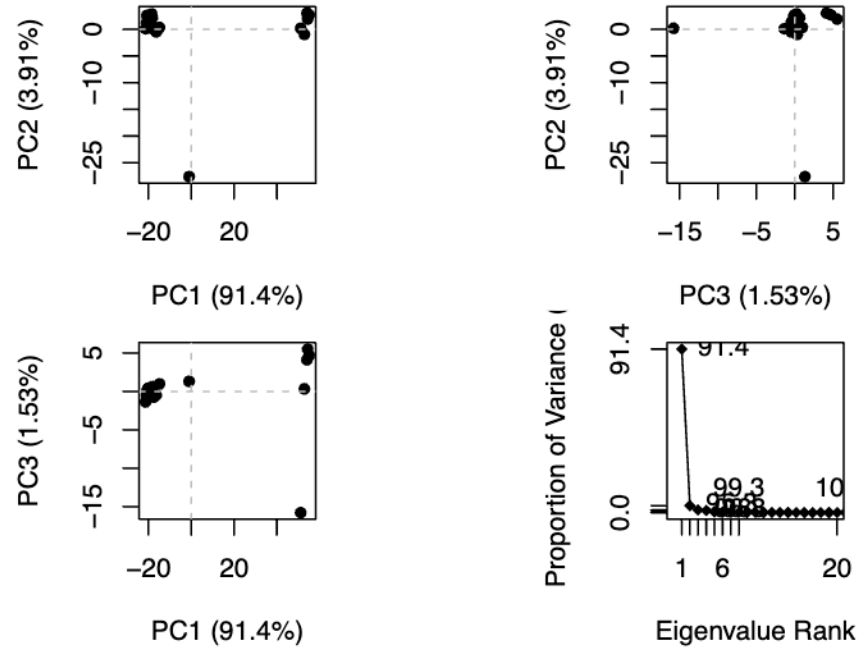
9R6U_A	Mattsson, J., et al. Biochemistry (2025)	NA	0.22790
9R71_A	Mattsson, J., et al. Biochemistry (2025)	0.19600	0.24400
8Q2B_A	Nam, K., et al. J Chem Inf Model (2024)	0.18320	0.22440
8RJ9_A	Nam, K., et al. Sci Adv (2024)	0.15190	0.20290
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500
4X8H_A	Kovermann, M., et al. Nat Commun (2015)	0.19610	0.28950
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al. Bioconj Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
8PVW_A	Rodriguez, J.A., et al. To be published	0.18590	0.23440
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
4NP6_A	Kim, Y., et al. To be published	0.18800	0.22200

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
8BQF_A	0.21882	P 2 21 21
4X8M_A	0.24630	C 1 2 1
6S36_A	0.15940	C 1 2 1
9R6U_A	0.19190	P 21 2 21
9R71_A	0.19300	P 21 21 2
8Q2B_A	0.18100	P 1 21 1
8RJ9_A	0.15010	P 21 21 2
6RZE_A	0.18190	C 1 2 1
4X8H_A	0.19140	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2
6HAM_A	0.20311	P 43
8PVW_A	0.18340	P 2 21 21
4K46_A	0.16730	P 21 21 21
4NP6_A	0.18600	P 43

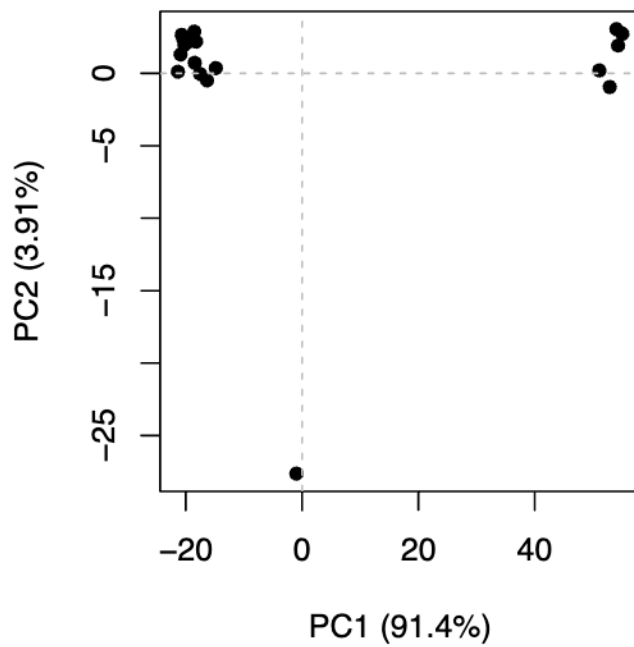
Principal Component Analysis

PCA of all this structural data (x, y and z atom coordinates):

```
# Perform PCA
pc <- pca(pdfs)
plot(pc)
```



```
# Create plot with PC 1 and 2
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

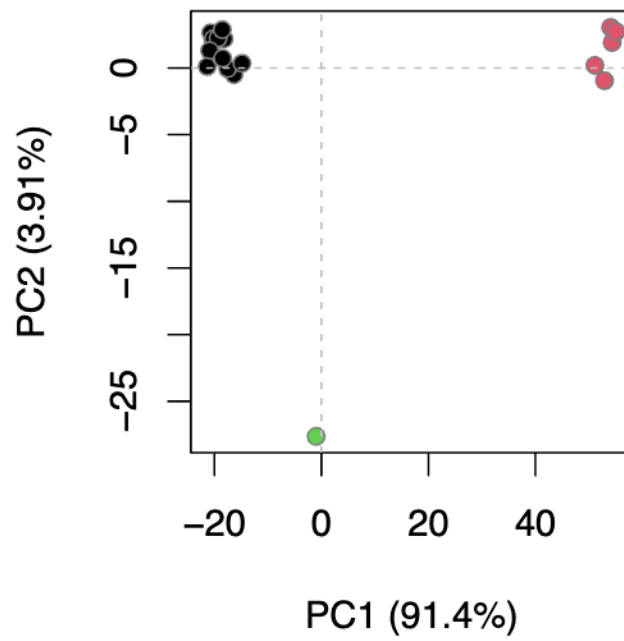
```
# Create interactive view of the PC1 captured structural differences
#| eval: !expr knitr::is_html_output()
#view.pca(pc)
```

```
# Calculate RMSD
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 182 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
# Visualize first principal component  
pc1 <- mktrj(pc, pc=1, file="pca.pdb")
```